Mudit Garg

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Research Interests

9 first-author papers (7 published) · SciX Library

I study how **massive black holes (MBHs) co-evolve with their host galaxies**, focusing on gas disks around MBH binaries (MBHBs) from sub-parsec scales to merger. I built the first pipeline from simulation-calibrated gas torques to Bayesian LISA inference and showed when gas and orbital eccentricity leave measurable imprints on gravitational waves (GWs). These imprints probe gas at micro-parsec scales that telescopes cannot resolve in distant binaries, and, if ignored, can mimic violations of general relativity. I now run 3D magnetohydrodynamic (MHD) simulations of turbulent circumbinary disks and have added my live post-Newtonian binary to a GPU code, to model both GW and EM signals, so LISA, Rubin, Roman, and COSI can jointly constrain the gas.

Education

08/'21 – 10/'25	Doctor of Philosophy, Faculty of Science **University of Zurich** Advisor: Prof. Lucio Mayer *Decoding the Sub-Parsec Scale Environment of LISA Massive Black Hole Binaries with Gravitational Waves*
09/'18 – 12/'20	Master of Science in Physics GPA: 5.87/6 **ETH Zurich** Thesis supervisor: Prof. Lavinia Heisenberg *Relativistic, ghost-free, and covariant hybrid model for MOND: $f(Q)$*
07/'14 – 06/'18	Bachelor of Technology in Engineering Physics GPA: 8.15/10 **IIT Delhi** Thesis supervisor: Prof. Ajit Kumar *Geodesics near a charged black hole in $(R \pm \mu^4/R)$ gravity*

Academic Experience

11/'25 – *10/'27*	SNSF Postdoc.Mobility Fellow (own grant and proposal) **New York University** Host: Prof. Andrew MacFadyen
02/'21 – 07/'21	Research Scholar **University of Zurich** Supervisor: Prof. Lucio Mayer

Fellowships & Awards

'25	SNSF Postdoc.Mobility Fellowship, CHF 136,000 (≈USD 165,000)
'20	MSc Physics with distinction, ETH Zurich

Leadership, Service, and Teaching

'26 –	Co-organizer, Astrophysics Seminars, New York University
'26 –	Mentor, LISA Early Career Scientists (LECS) mentoring program
'25 –	Referee for ApJ, PRD
'24 –	Outreach: video interviews for the LISA Consortium ('24) and *The Minds in Motion* ('26); Three Minute Thesis talk ('25)

'22 –	Contributor, LISA Consortium projects: Disc-IMRI hydro code comparison (paper); astrophysical source parameter space for waveforms; MBHB section (§2.2) of the in-progress update of the LISA Astrophysics white paper
'21 –	LISA Consortium member, core member since '25; Astrophysics, Waveform, LISA Data Challenge, and LECS groups
'24	Informal advisor to a junior PhD student on LISA detectability, University of Zurich
'23 – '25	Sole organizer, "GWs, BHs, and Compact Binaries" Seminars, University of Zurich
'21 – '25	Teaching assistant, University of Zurich: *Theoretical Astrophysics*, *Computational Methods for Radiative Transfer*, and others

Selected Talks 9 seminars, 1 colloquium, 16 conference talks (4 invited†, 1 panel) · full list

Seminars & colloquia: GRAPPA Colloquium, Amsterdam ('26); THEA Seminar, Columbia ('26); CfA High Energy Seminar, Harvard & Smithsonian ('26); Special TAPIR Seminar, Caltech ('25); Cosmology Seminar, MPA Garching ('24); DAMTP GR Seminar, Cambridge ('24).

Conferences: DYNAMIX†, Cambridge ('25); GW Probes of BH Environments†, Trieste ('25); Frontiers of Astrophysical Black Holes†, Sexten ('25); New Ideas on the Origin of BH Mergers†, NBI Copenhagen ('24); Panelist and short talk, eccentric compact binaries, Corfu ('26); LISA Symposium, College Park ('26).

Informal: CCA SPA ('24, '26) and GW ('26) groups, and Wednesday coffee talk ('26); MIT Kavli ('24, '26); IAS-Princeton GW group ('26).

Research Visits

- CCA, Flatiron Institute: regular visitor, MHD and `Pasta` work with Y.-F. Jiang ('25–)
- Niels Bohr International Academy, Copenhagen ('25; host J. Samsing)
- Max Planck Institute for Gravitational Physics (AEI), Potsdam ('23; host J. Gair)

Codes & Computing C++ (Kokkos), Python, Mathematica

- `Pasta` (GPU radiation-MHD code by Jiang et al., in prep.): wrote the live post-Newtonian (PN) binary setup (≈1,700 lines: 1–2.5PN dynamics, gas-gravity back-reaction, standard and torque-free sink accretion, target-temperature cooling), validated against PN theory and my `Athena++` MHD runs, found the multi-GPU MHD bottleneck, and added faster POSIX output.
- `Athena++`: matched 3D MHD and α-viscosity hydrodynamic (HD) simulations of circumbinary disks (1.2M CPU core-hours); implemented a locally isothermal equation of state.
- `GIZMO`: HD simulations of live PN MBHBs in circumbinary disks (350k CPU core-hours).
- `lisabeta`: developed the environmental-effect waveform modules and included the eccentric TaylorF2 model (Moore et al. 2016) as `TF2Ecc`, with S. Tiwari; full LISA response and Bayesian parameter estimation with `ptmcmc` and `Eryn` (RJMCMC).
- **Allocations:** collaborator on the Empire AI GPU request (PI A. MacFadyen, in preparation).

Mudit Garg: Publications

10 refereed (7 first-author) · 3 under review (2 first-author) · 1 preprint
3 first-author papers since the PhD
465 citations (188 to first-author papers), *h*-index 10 (SciX, Oct 2026)

First Author

Under review

14. **Mudit Garg**, Yan-Fei Jiang
MRI Turbulence versus α-Viscosity in Circumbinary Disks: Implications for Gravitational-Wave Dephasing and Accretion Variability, 2026, arXiv:2610.17212 (under review at ApJ)

13. **Mudit Garg**, Alessia Franchini, Alessandro Lupi
Gas-induced gravitational-wave dephasing and accretion periodicities of live post-Newtonian massive black hole binaries: warm disk, 2026, arXiv:2606.20787 (under review at ApJ)

Published

12. **Mudit Garg**, Lucio Mayer, Yinhao Wu, Yacine Ali-Haïmoud, Douglas N. C. Lin
Chaotic Migration of LISA Extreme Mass Ratio Inspirals in a Turbulent Accretion Disk: Effect on Waveform Dephasing, 2026, ApJL, 1009(1), L8, DOI: 10.3847/2041-8213/aea0ca

11. **Mudit Garg**, Alessia Franchini, Alessandro Lupi, Matteo Bonetti, Lucio Mayer
Gas-induced Perturbations on the Gravitational Wave Inspiral of Live Post-Newtonian LISA Massive Black Hole Binaries, 2025, ApJ, 993(1), 145, DOI: 10.3847/1538-4357/ae10b4

10. **Mudit Garg**, Laura Sberna, Lorenzo Speri, Francisco Duque, Jonathan Gair
Systematics in tests of general relativity using LISA massive black hole binaries, 2024, MNRAS, 535(4), 3283, DOI: 10.1093/mnras/stae2605

9. **Mudit Garg**, Christopher Tiede, Daniel J. D'Orazio
Accretion-mediated spin-eccentricity correlations in LISA massive black hole binaries, 2024, MNRAS, 534(4), 3705, DOI: 10.1093/mnras/stae2357

8. **Mudit Garg**, Andrea Derdzinski, Shubhanshu Tiwari, Jonathan Gair, Lucio Mayer
Measuring eccentricity and gas-induced perturbation from gravitational waves of LISA massive black hole binaries, 2024, MNRAS, 532(4), 4060, DOI: 10.1093/mnras/stae1764

7. **Mudit Garg**, Shubhanshu Tiwari, Andrea Derdzinski, John G. Baker, Sylvain Marsat, Lucio Mayer
The minimum measurable eccentricity from gravitational waves of LISA massive black hole binaries, 2024, MNRAS, 528(3), 4176, DOI: 10.1093/mnras/stad3477

6. **Mudit Garg**, Andrea Derdzinski, Lorenz Zwick, Pedro R. Capelo, Lucio Mayer
The imprint of gas on gravitational waves from LISA intermediate-mass black hole binaries, 2022, MNRAS, 517(1), 1339, DOI: 10.1093/mnras/stac2711

Co-Author

Under review

5. Andrea Derdzinski et al. (**Mudit Garg**: 14th of 19 authors)
The LISA Astrophysics "Disc-IMRI" Code Comparison Project: Intermediate-Mass-Ratio Binaries in AGN-Like Discs, 2025, arXiv:2512.10893 (under review at Open J. Astrophys.)
Contribution: text, figures, literature review

Published

4. En-Kun Li et al. (**Mudit Garg**: 21st of 53 authors)
Gravitational wave astronomy with TianQin, 2025, Rep. Prog. Phys., 88(5), 056901, DOI: 10.1088/1361-6633/adc9be
Contribution: sole author of Sec. IV.B.2, "Dephasing in gas"

3. Lorenz Zwick, Andrea Derdzinski, **Mudit Garg**, Pedro R. Capelo, Lucio Mayer
Dirty waveforms: multiband harmonic content of gas-embedded gravitational wave sources, 2022, MNRAS, 511(4), 6143, DOI: 10.1093/mnras/stac299
Contribution: idea development, one figure, editing

2. Fabio D'Ambrosio, **Mudit Garg**, Lavinia Heisenberg (alphabetical order)
Non-linear extension of non-metricity scalar for MOND, 2020, Phys. Lett. B, 811, 135970, DOI: 10.1016/j.physletb.2020.135970
Contribution: derived the main result as part of my master's thesis; writing

Preprint

1. Fabio D'Ambrosio, **Mudit Garg**, Lavinia Heisenberg, Stefan Zentarra (alphabetical order)
ADM formulation and Hamiltonian analysis of Coincident General Relativity, 2020, arXiv:2007.03261
Contribution: derived the main result as part of my master's thesis; writing

Further Research Outputs

- **PhD thesis: Mudit Garg**, *Decoding the Sub-Parsec Scale Environment of LISA Massive Black Hole Binaries with Gravitational Waves*, 2025, University of Zurich, DOI: 10.5167/uzh-281206

- **Software:** `TF2Ecc` (2024), the eccentric TaylorF2 waveform model (Moore et al. 2016), included in the LISA inference code `lisabeta` by **Mudit Garg** with Shubhanshu Tiwari (used in first-author papers 7, 8, 10 above), gitlab.in2p3.fr/marsat/lisabeta